



COURSE SYLLABUS						
Course Code	INTE 202	Course Title	Integrative Programming and Technologies 1		Course Credit	5
Course Description		Integrative Programming and Technologies explores the principles and practices of combining multiple programming languages, tools, and technologies to create unified and functional systems. The course emphasizes the integration of front-end and back-end components, database connectivity, and web or software services to develop efficient, scalable, and user-centered applications. Students will learn how to design, implement, and evaluate integrated solutions that address real-world problems. Through project-based learning, the course fosters problem-solving, collaboration, and the responsible use of technology in modern development environments.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				

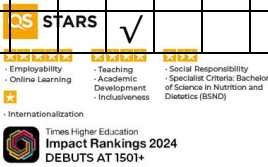


INSTITUTIONAL LEARNING OUTCOMES (ILO)	As a polytechnic state university, PUP shall develop its students to possess: 1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.
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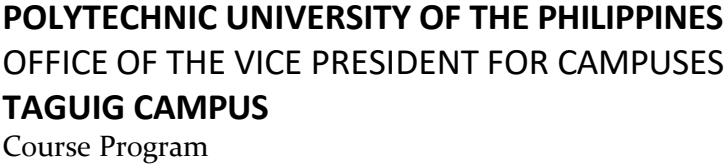
	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10. Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		✓			✓					
			✓							
	Apply critical and creative thinking in analyzing problems and developing web-based solutions using PHP, HTML, MySQL, and API technologies.				✓					
	Communicate effectively in both oral and written forms to present system requirements, design documentation, and technical solutions to diverse stakeholders.		✓							
	Demonstrate strong service orientation by developing applications that address the needs			✓						✓





	of communities, industries, and organizations.									
	Utilize and develop technology responsibly through the application of secure, efficient, and sustainable web technologies and programming practices.			√			√			



	Engage in lifelong learning by keeping pace with emerging programming languages, frameworks, and development trends, adapting skills to meet evolving industry demands.				√			√
	Work collaboratively and ethically in multidisciplinary teams to design, develop, and deploy software applications.					√	√	√
	Employ problem-solving and research skills in designing algorithms, managing databases, and integrating APIs to create functional and innovative systems.	√		√				

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain and apply basic web development concepts including HTML structure, PHP syntax, and MySQL operations.	√						√	
	Design and develop functional web pages using HTML forms, tables, and semantic elements.	√	√						
	Develop PHP programs that handle input/output, decision-making, loops, and functions for solving programming tasks.	√						√	
	Integrate PHP with MySQL to perform CRUD operations (Create, Read, Update, Delete) and manage databases effectively.	√			√			√	
	Implement session and authentication mechanisms in PHP to ensure secure access and responsible handling of user data.				√		√	√	
	Consume and create simple RESTful APIs using PHP and MySQL to exchange JSON data with external applications.				√			√	
	Work collaboratively on a mini-project that applies web programming, database integration, and API development to address a real-world problem.			√			√		
	Demonstrate effective communication by preparing documentation and presenting the design and functionality of their web applications.		√				√		
	Apply problem-solving and debugging strategies to identify and fix errors in PHP, MySQL, and API implementations.	√				√			√




General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU





	Exhibit professional and ethical behavior in coding practices, teamwork, and handling of data.				✓		✓	✓	
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OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs										
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4	5	6	7	8	9	10	
					Face-to-Face	FLTAs											
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X										
2	- Explain the basic structure of the web - Set up a working development environment - Create and run a basic webpage - Develop foundational understanding	Introduction to Web Development - Overview of web technologies (frontend vs backend) - Basics of client-server architecture - Setting up development environment (XAMPP/LAMP/WAMP or Docker) - Writing the first	Lecture-Discussion, Demonstration		Short quiz on web basics, submission of first HTML page.		X									X	



2025

7th in the Philippines
92nd in Southeast Asia
53rd in Africa



CS STARS

4★



Sustainability

4★



Social Responsibility

4★



		HTML page																	
3	- Understand HTML document structure - Use common HTML tags - Build interactive forms - Apply semantic HTML principles	HTML Fundamentals - Structure of an HTML document - Common tags (headings, paragraphs, links, images, lists) - Forms and input elements - Introduction to semantic HTML	Lecture, Hands-On Lab		Lab exercise (create a profile webpage with forms and tables)		X	X										X	
4 - 5	- Understand PHP syntax and fundamentals - Apply operators and expressions - Implement control structures - Develop simple PHP programs	Introduction to PHP - PHP syntax, variables, and data types - Basic operators and expressions - Control structures (if, switch, loops) - Writing a simple PHP script	Lecture, Coding Demonstration, Guided Practice		Seatwork (PHP program with conditionals and loops).		X		X									X	
6	- Create and use functions in PHP - Work with form data transmission methods - Handle form inputs programmatically - Implement basic validation and sanitization	PHP Functions and Forms - Defining and using functions - Working with GET and POST methods - Handling form inputs with PHP - Basic input validation and sanitization	Lecture-Discussion, Problem-Based Learning		Lab exercise (build a login form simulation with validation).				X									X	X
7 - 8	Understand	Introduction to	Lecture,		Quiz (SQL syntax)													X	



	relational database concepts • Work with MySQL database structures • Apply basic SQL commands • Integrate MySQL with PHP applications	MySQL • Relational databases overview • Introduction to MySQL (tables, rows, columns) • Basic SQL commands (CREATE, INSERT, SELECT, UPDATE, DELETE) • Connecting PHP with MySQL (mysqli / PDO)	Hands-On Lab, Guided Discovery		lab output (student table with insert and select queries).												
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Times Higher Education

Impact Rankings 2024

DEBUTS AT 1501+



9		MIDTERM WRITTEN EXAMINATION														
10	- Retrieve and display database content - Insert data through web forms - Modify and remove records - Handle errors and debug applications	PHP + MySQL Integration - Fetching and displaying data with PHP - Inserting data into database via forms - Updating and deleting records - Error handling and debugging	Demonstration, Pair Programming		Lab exercise (simple student information system with add/view/delete).					X						X
11	- Understand and apply PHP state management - Build basic authentication systems - Store and manage user data securely - Implement password hashing for security	State Management in PHP - Understanding sessions and cookies - User authentication basics (login/logout system) - Storing user data securely - Password hashing with PHP	Lecture, Guided Practice, Discussion		Lab exercise (secure login page with sessions).						X					X
12	Organize PHP code for	PHP and Web Application	Project-		Group lab output		2025		X	X	X	X			X	X



	- scalability - Develop data-driven applications with CRUD functionality - Understand MVC as a design pattern	Development - Structuring code for small web apps - CRUD operations in PHP + MySQL (Mini Project: Student Record System) - Introduction to MVC (conceptual, not full framework)	Based Learning , Group Work		(working CRUD app with authentication).													
13	- Understand the concept of APIs in web development - Consume third-party APIs using PHP - Work with JSON data in PHP	Introduction to APIs - What is an API? (REST concepts, JSON, endpoints) - Consuming third-party APIs in PHP (example: weather or news API) - Handling JSON data in PHP	Lecture, Demonstration, Hands-On		Lab exercise (fetch and display API data in a webpage).		X					X			X			
14	Design and build a simple REST API in PHP	Creating Your Own API Designing a simple REST API	Guided Practice, Problem-		Lab exercise (create API that returns JSON for								X		X	X		



	• Send and receive JSON data • Implement CRUD functionality through API endpoints • Test and validate APIs effectively	• in PHP • Sending and receiving JSON data • Implementing CRUD operations via API endpoints • Testing APIs with Postman •	Based Learning		student records).												
15	FINAL WRITTEN EXAMINATION																



LIST OF REFERENCES

BOOKS:

Duckett, J. (2011). HTML & CSS: Design and Build Websites. Wiley.
Nixon, R. (2014). Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 (4th Edition). O'Reilly Media.
Ullman, L. (2017). PHP and MySQL for Dynamic Web Sites: Visual QuickPro Guide (5th Edition). Peachpit Press.
Meloni, J. C. (2018). PHP, MySQL, & JavaScript All in One. Sams Publishing.

WEBSITE:

W3Schools – HTML Tutorial
<https://www.w3schools.com/html/>

W3Schools – PHP Tutorial
<https://www.w3schools.com/php/>

MySQL Official Documentation
<https://dev.mysql.com/doc/>

PHP Manual – Official Documentation
<https://www.php.net/manual/en/>

MDN Web Docs – Web Development Guides
<https://developer.mozilla.org/>

Postman API Learning Center
<https://learning.postman.com/>



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a





	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.											
COURSE REQUIREMENT/S	GRADING SYSTEM											
- Students are required to complete hands-on exercises, summative quizzes, and a short reflection paper to demonstrate understanding of integration concepts. A midterm project proposal will outline their planned web-based system, while a final project will involve the full implementation and presentation of an integrated web application with documentation. Active participation in class discussions and activities is also expected. The final grade will be based on class standing (70%) and major examinations (30%), with the final grade computed as the average of the midterm and final term grades.	<table><tr><th>Mid Term</th><th>Final Term</th></tr><tr><td> Class Standing70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Midterm Examinations30% 100% </td><td> Class Standing70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Final Examinations30% 100% </td></tr><tr><td colspan="2"></td></tr><tr><td colspan="2">Final Grade = (Midterm Grade + Final Term Grade) / 2</td></tr><tr><td colspan="2"></td></tr></table>	Mid Term	Final Term	Class Standing70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Midterm Examinations30% 100%	Class Standing70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Final Examinations30% 100%			Final Grade = (Midterm Grade + Final Term Grade) / 2				
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	Final Grade = (Midterm Grade + Final Term Grade) / 2											
CLASS INFORMATION	FACULTY INFORMATION											
Course, Year and Section: BSIT 2-1	Name of Faculty: Ronald Joy G. Tengco											
Day and Time: W/W 04:00PM-07:00PM/07:00PM-09:00PM	Consultation Time:											
Room:	Office Tel. No./ Mobile Phone No. 09190963824											
School Year: 2025-2026 Semester: Second	E-Mail Address: rjgtengco@pup.edu.ph											



Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
 RONALD JOY G. TENGCO Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	 DR. RHYAN V. MOLINAR Director	DR. ARMANDO A. TORRES Vice President for Campuses